Multi-Backbone SED Ensembles with Snippet-Bank Matching for Indic Noise Event Detection

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Abstract—We describe our Track 1 system for the Datathon@IndoML 2026 noise event detection challenge: localising class-agnostic onset/offset pairs of noise events in Indic speech from the Vaani corpus, scored by event F1 plus segment Dice [10]. The system combines (i) an ensemble of twelve frame-level sound event detectors built on five pretrained encoders (ATST-F, BEATs, WavLM-base+/large, w2v-BERT 2.0), each fused with a mel-CNN and a BiGRU emitting 24 presence and boundary tracks; (ii) a snippet bank: the organizers' synthetic mixtures reuse sample-exact noise snippets, which we isolate from validation pairs and detect with calibrated matched filtering; and (iii) a routing decoder, tuned under test-mixture weighting, that merges bank and neural evidence per clip. Our final submission (Codabench 995227) reaches event F1 0.7711 and segment Dice 0.8232, a combined score of 1.5943 (rank 6), up from 1.2129 for our Phase 1 entry.

Index Terms—sound event detection, ensembles, self-supervised encoders, matched filtering, Indic speech

I. Training Data and Preprocessing

A. Data sources

Table I lists the data. No external datasets are used beyond the pretrained encoders of Table II. The IndoML validation set is split into five folds; fold 4 (479 clips: 344 natural, 135 synthetic) is the monitor and selection hold-out and is never trained on.

B. Targets and synthetic mixtures

Each clip yields 24 channels at a 20 ms frame rate: union presence, seven category presence tracks, and union plus per-category onset/offset tracks (Gaussian boundary targets). The organizers' synthetic clips are built as noisy = clean + $\sum_i g_i$ snippet$_i$ with labels exact to the sample. We regenerate such mixtures on the fly, using event counts, gaps (≥200 ms), gains and durations measured on the 695 validation synthetic clips. Noise comes from the snippet bank and, for widened members, isolated Gold event spans; speech comes from validation clean references (held-out folds excluded) and Gold/Silver speech regions at least 50 ms from any event.

TABLE I
Training data. Vaani tiers follow the dataset's annotation quality.

Source	Size	Use
Vaani [1] Gold (verified timestamps)	11,111 seg., 21.8 h	frame targets; noise events and speech for mixtures
Vaani Silver (unverified timestamps)	61,642 seg., 100.3 h	frame targets (down-weighted)
Vaani Bronze (tags only)	17,884 seg., 32.4 h	multiple-instance (MIL) clip head
IndoML validation, natural, folds 0–3	1,462 clips	extra frame targets for six members
IndoML validation, synthetic	695 pairs	snippet bank (206 snippets); clean speech for mixtures

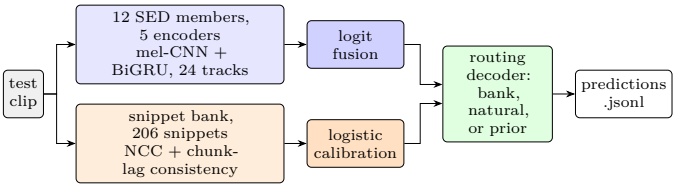


Fig. 1. System overview. Fused neural probabilities and calibrated bank matches meet in a per-clip routing decoder; every clip receives at least one event.

C. Batches and augmentation

Batches mix the sources in fixed proportions (e.g. Gold 0.40, Silver 0.27, Bronze 0.05, synthetic 0.28). Crops are event-biased: 80% contain a whole event and 10% are hard negatives, with 10 s crops (4 s for the breath specialist). Augmentations are random gain, label-preserving whole-frame time shifts, label-union mixup within a tier, Filter-Augment [7] band gains, and optional Freq-MixStyle.

II. Proposed Method

A. Neural SED members

Each member (Fig. 1) fuses a pretrained encoder (Table II) with a mel-CNN (512-dimensional projection, part dropout 0.1); a masked 2-layer BiGRU (hidden 256) then outputs the 24 tracks, alongside a linear-softmax MIL head with seven clip logits. Table III lists the twelve members. Our Phase 1 WavLM-base+ detectors s9000/s1600 instead use learned layer weights, a BiGRU, 1+7+64 tracks, boundary heatmaps and attention-MIL, and are trained on Vaani data only.

Losses. Presence uses BCE or focal loss per track plus a soft-Dice loss on the union track; boundaries use a RED-style focal loss; Bronze clips contribute the MIL loss only. Tier weights (presence/boundary) are Gold 1.0/1.0, Silver 0.5/0.1, synthetic 1.0/1.0 and validation natural 1.5/1.5, with Silver background frames weighted ×0.5.

TABLE II
Pretrained encoders.

Encoder	Version and source	Members
ATST-F strong [3]	PretrainedSED release v0.0.1 [2] (ATST-F_strong_1.pt)	m1, m1v, m1g, m6, m7, m8
BEATs strong [4]	PretrainedSED release v0.0.1 [2]	m2
WavLM base+ [5]	microsoft/wavlm-base-plus (HF)	m4, s9000, s1600
WavLM large [5]	microsoft/wavlm-large (HF)	m3
w2v-BERT 2.0 [6]	facebook/w2v-bert-2.0 (HF)	m5

TABLE III
Ensemble members. "Val" marks training on validation natural clips (folds 0–3). Fold-4 is each member's best training-time monitor score (combined, threshold 0.5, no bank; "–": not logged). w is the final fusion weight.

Member	Encoder, variant	Val	Fold 4 (step)	w
m1	ATST-F	–	1.383 (18k)	.028
m4	WavLM-b+, breath specialist	–	1.395 (12k)	.252
m7	ATST-F, seed 77, widened mix	–	–	.070
m8	ATST-F, all-layer mixing	–	–	.168
s9000	WavLM-b+ (Phase 1)	–	–	.112
s1600	WavLM-b+ (Phase 1)	–	–	.070
m1v	ATST-F	✓	1.382 (9k)	.050
m1g	ATST-F, continued on Gold	✓	1.391 (9k)	.050
m6	ATST-F, seed 91, widened mix	✓	–	.050
m2	BEATs, widened mix	✓	1.341 (7.8k)	.050
m3	WavLM-large, widened mix	✓	1.339 (9k)	.050
m5	w2v-BERT 2.0, widened mix	✓	1.410 (10k)	.050

Training. Stage 1 trains the new modules with the encoder frozen (1,000–1,500 steps, learning rate 10^{-3}); stage 2 unfreezes it (learning rate 1–2·10^{-5}, layer-wise decay 0.85–0.9). We use AdamW, bf16, gradient clipping and EMA weights [8], with a cosine schedule fitted to a wall-clock deadline from measured step times. A fold-4 monitor keeps the two best checkpoints, student or EMA (Fig. 2). Inference uses 10 s windows with a 5 s hop, triangular overlap weights (floor 0.3) and time-shift TTA over $\{0, \pm 20\}$ ms.

B. Snippet bank

Subtracting clean from noisy audio at the labelled spans of the 695 validation synthetic pairs isolates 2,213 events, which deduplicate to 206 distinct snippets. For every snippet, normalised cross-correlation (NCC) is computed against each clip via FFT, keeping up to three peaks with NCC $\geq$ 0.35; edge sub-templates (first or last 250 ms) catch events truncated by the clip boundary. Each candidate is described by its NCC, a chunk-lag consistency score (whether every 50 ms chunk peaks at the same lag, a property of sample-exact copies but not of natural look-alikes), the implied SNR, the snippet duration and an edge flag. A logistic model calibrated leave-fold-out maps these to a match probability, and greedy non-maximum suppression yields the bank events.

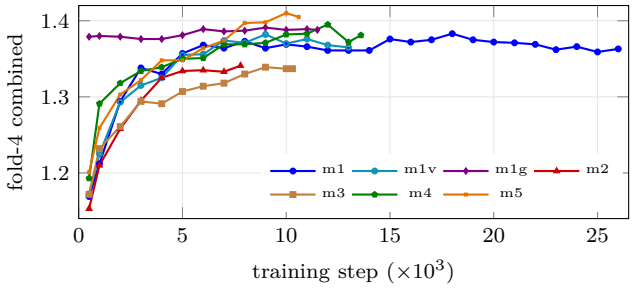


Fig. 2. Fold-4 monitor score (best of student/EMA) during training, from the training logs, for the seven members trained on our cluster (names as in Table III; m1g continues from m1). Each released checkpoint is the step with the best monitor score.

TABLE IV
Combined score (event F1 + segment Dice) on validation and on the Codabench test set for our Phase 1 entry and every Phase 2 submission. Validation scores of ensembles with validation-trained members include their training folds; fold 4 is held out.

ID	Configuration	Val. (2,501)	Fold 4 (479)	Test
938975	s9000+s1600 (Phase 1)	1.134	–	1.2129
995013	interim, clean members	1.468	1.466	1.5803
995109	all 12, fold-4 tuned, test-mix	1.541	1.512	1.5854
995114	all 12, fold-4 tuned	1.541	1.512	1.5857
995138	clean members, test-mix	1.482	1.465	1.5875
995227	clean + 30% others	1.533	1.480	1.5943

C. Fusion, routing and decoding

Member caches are fused in the logit domain with the weights of Table III (configuration clean_plus_a30: 70% on the six members trained without validation natural clips, 5% on each of the other six). The highest calibrated bank probability per clip (its synthetic score) selects one of three paths: bank, where bank events above 0.56 confidence are merged with neural events above 0.64 (margin 0.18, gap 200 ms, at most five events); natural, a category-aware hysteresis decoder (thresholds 0.85/0.61, median filter of 5 frames, minimum 40 ms, merge gap 160 ms, boundary padding $-20/-30$ ms) that splits blobs whose category tracks disagree, so overlapping events of different types become separate events; and prior, a fallback so that no clip is left empty. All thresholds and fusion weights are searched with Optuna TPE [9] on validation, with synthetic clips weighted $\times 1.57$ to match the test mixture of 2,080 synthetic and 3,437 natural clips; the tuner reports a 5-fold CV estimate next to the full-validation score.

III. Results and Conclusion

Tables IV–V and Fig. 3 summarise the results.

- Final entry (995227): event F1 0.7711 and segment Dice 0.8232 on the 5,517 test clips, combined 1.5943. It routes 1,974 clips through the bank (the test set

TABLE V
Final configuration on validation by clip type (test-mix weighting). The bank's snippets come from the synthetic validation clips themselves, so their synthetic score is an in-sample upper bound.

Subset	Clips	Event F1	Seg. Dice	Combined
Natural	1,806	0.429	0.721	1.150
Synthetic (in-sample bank)	695	0.999	0.996	1.995
All, test-mix weighted	2,501	0.749	0.824	1.574
5-fold CV of the full configuration: 1.572 ± 0.021 (mean ± std)				

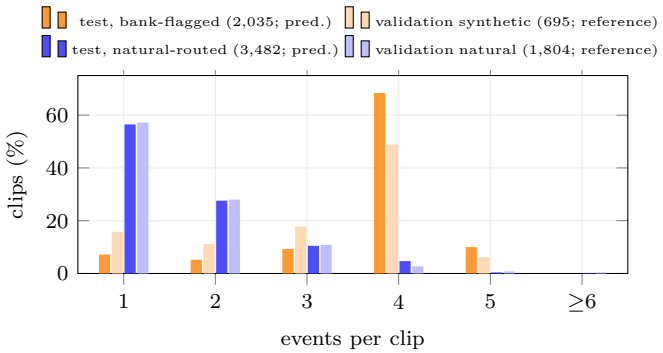


Fig. 3. Events per clip in our final test predictions, split by bank flag, against the validation reference annotations. Natural-routed test predictions closely follow the natural reference distribution.

has 2,080 synthetic clips), 3,482 through the natural decoder and 61 through the prior, emitting 13,268 events.

- The bank transfers to the test set. Event F1 rose from 0.47 for our Phase 1 entry to 0.77.
- Natural clips are the hard half. On validation, natural clips reach only 1.150 combined (Table V); single members reach 1.34–1.41 on fold 4 (Table III), and fusion lifts the ensemble to 1.48 there.
- Members trained without validation natural clips generalise best. clean_full_tm scores lower than final_f4_tm on validation but higher on test; re-adding the other members at 30% total weight gave the best test score.
- Category-aware decoding matters. About 30% of consecutive natural event pairs overlap; splitting fused blobs by category track recovers them as separate events.

Novelty. (i) Calibrated, consistency-checked matched filtering that exploits the reuse structure of the synthetic mixtures; (ii) a generator that reproduces the organizers' mixtures with sample-exact labels for training; (iii) a heterogeneous ensemble over five pretrained encoders, fused in logits under a held-out-aware weighting; and (iv) test-mixture-weighted decoder tuning with routing.

Conclusion. Heterogeneous pretrained SED members combined with an explicit model of how the synthetic data was built give a strong, well-calibrated detector. Natural clips remain the harder half; better boundary modelling of overlapping natural events is the main avenue for improvement.

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